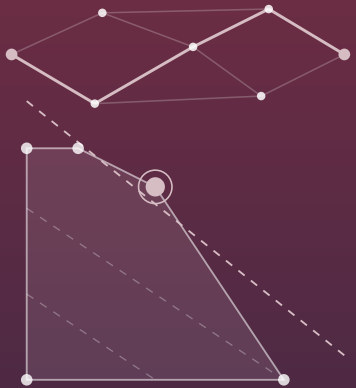
Operations Research

Lecture 4 · Semester 5



المدرسة الوطنية العليا لتكنولوجيا الأنظمة المستقلة

NHSAST

NATIONAL HIGHER SCHOOL OF AUTONOMOUS SYSTEMS TECHNOLOGY

Starting the Simplex Method

The two-phase method, infeasible problems and degeneracy

NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

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Operations Research (O.R.) · Semester 5 · 2026–2027

Today's journey: six parts



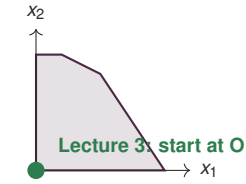
1. The origin is not always a corner
2. Phase 1: find a first corner
3. Phase 2, and the two-phase method
4. Example: a minimum order
5. No feasible plan, and special cases
6. Degeneracy, and the complete method

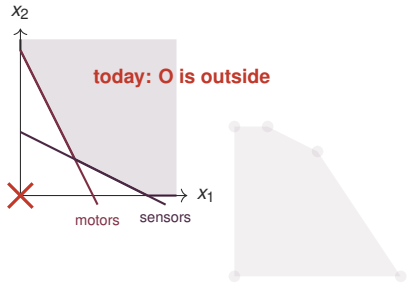
Goal solve any linear program with the simplex method.

Where we are: the start of Lecture 3

- Lecture 3: $\max\ c^\top x,\ Ax \le b,\ x \ge 0$, with $b \ge 0$.
- The slacks gave the first corner for free: $x = 0$, $s = b \ge 0$.
- But many problems have "$\ge$" constraints (a need, a minimum) or "$=$" constraints (exactly).
- Then the origin is not feasible: no corner to start from.

Today use the simplex method itself to find a first corner.





Part 1 of 6 · minutes 5–16

The origin is not always a corner

1. **Origin not a corner**
2. Phase 1
3. Phase 2
4. Example: minimum order
5. No feasible plan
6. Degeneracy, whole method

In this part, you learn to

- see why the origin can be outside the feasible set;
- prepare any problem with slack, surplus and artificial variables (Rule 4.1);
- find the first basic variables.

The problem: the origin is not a corner

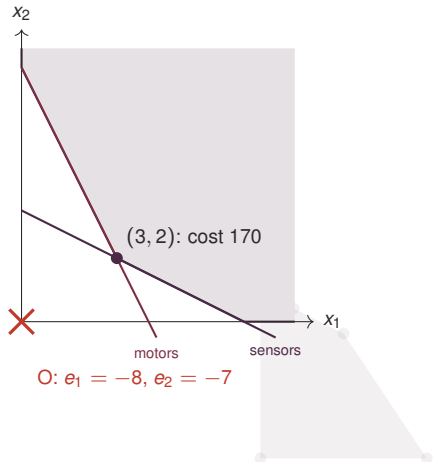
Spare-part boxes (Lecture 2):

$$\begin{aligned} \min\ & z = 30x_1 + 40x_2 \\ \text{s.t. } & 2x_1 + x_2 \geq 8 \quad \text{(motors)} \\ & x_1 + 2x_2 \geq 7 \quad \text{(sensors)}, \quad x \geq 0 \end{aligned}$$

Standard form, with surplus variables (Rule 1.6):

$$2x_1 + x_2 - e_1 = 8, \qquad x_1 + 2x_2 - e_2 = 7.$$

At the origin: $e_1 = -8$, $e_2 = -7$: not feasible.
The surpluses cannot be the first basic variables.



Rule 4.1: prepare the equations

Step 1. If a right-hand side is negative, multiply the constraint by -1 ("$\leq$" becomes "$\geq$").

Step 2. Line by line (now $b \geq 0$):

Constraint	we write	basic at the start
$\cdots \leq b$	$\cdots + s = b$ (slack)	$s = b$
$\cdots \geq b$	$\cdots - e + a = b$ (surplus, artificial)	$a = b$
$\cdots = b$	$\cdots + a = b$ (artificial)	$a = b$

Result each line has its own starting variable, equal to $b \geq 0$: a first basis.

Spare-part boxes: $2x_1 + x_2 - e_1 + a_1 = 8, \quad x_1 + 2x_2 - e_2 + a_2 = 7.$

Your turn: prepare the equations

$$\min\ x_1 + x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \geq 4, \quad x_1 - x_2 = 1, \quad x_1 \leq 5, \quad x_1, x_2 \geq 0.$$

Your turn · 2 min Which variables do you add to each line? Which variables are basic at the start?

Answer

$$x_1 + 2x_2 - e_1 + a_1 = 4 \quad (\text{“}\geq\text{”: surplus and artificial})$$

$$x_1 - x_2 + a_2 = 1 \quad (\text{“}=\text{”: artificial})$$

$$x_1 + s_3 = 5 \quad (\text{“}\leq\text{”: slack only})$$

Basic at the start: $a_1 = 4$, $a_2 = 1$, $s_3 = 5$.

Part 2 of 6 · minutes 16–34

Phase 1: find a first corner

1 Origin not a corner — 2 **Phase 1** — 3 Phase 2 — 4 Example: minimum order — 5 No feasible plan — 6 Degeneracy, whole method

In this part, you learn to

- read an artificial variable as a missing quantity;
- write the Phase 1 objective W and clean its line (Rule 4.2);
- run Phase 1 and read its end.

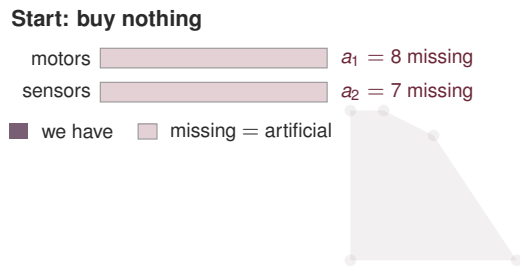
Definition 4.1: artificial variables

Definition 4.1

An artificial variable $a_i \geq 0$ is added to a "$\geq$" or "$=$" line only to have a first basis. It has no meaning in the real problem: a plan is real only when all artificial variables are 0.

$$2x_1 + x_2 - e_1 + a_1 = 8, \quad x_1 + 2x_2 - e_2 + a_2 = 7.$$

- Start: $x_1 = x_2 = e_1 = e_2 = 0$, so $a_1 = 8$, $a_2 = 7$.
- A useful reading: $a_1 =$ motors missing, $a_2 =$ sensors missing.
- The start is a fake plan. Our job: push a_1 and a_2 down to 0.

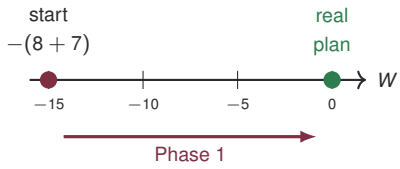


Phase 1: make the missing quantities zero

Phase 1 problem

max $W = -(a_1 + a_2)$ subject to the prepared equations, all variables ≥ 0.

- $W \leq 0$ always, and $W = 0$ exactly when $a_1 = a_2 = 0$.
- The start ($a = b$) is a corner of this problem: the simplex method of Lecture 3 applies.



Key fact 4.1

The real problem has a feasible plan $\iff$ the best value of W is 0.

Rule 4.2: clean the W-line before the first pivot

Start	x_1	x_2	e_1	e_2	a_1	a_2	RHS
a_1	2	1	-1	0	1	0	8
a_2	1	2	0	-1	0	1	7
W	0	0	0	0	1	1	0

must be 0!

W	0	0	0	0	1	1	0
$-L_1$	-2	-1	1	0	-1	0	-8
$-L_2$	-1	-2	0	1	0	-1	-7
W_{new}	-3	-3	1	1	0	0	-15

- $W = -a_1 - a_2$: line $W + a_1 + a_2 = 0$.
- a_1, a_2 are basic: their columns need 0 in the objective line.
- So subtract the lines of a_1 and a_2.
- Check: $-15 = -(8 + 7)$. ✓

Rule 4.2

Before the first pivot of a phase: 0 under every basic variable in the objective line.

Phase 1, pivot 1: from O to $(4, 0)$

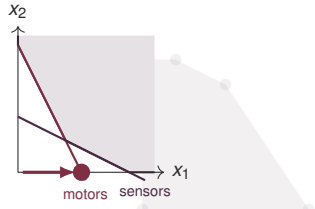
enters ↓ (tie: x_1)

T0	x_1	x_2	e_1	e_2	a_1	a_2	RHS	ratio
leaves a_1	2	1	-1	0	1	0	8	$8/2 = 4$
a_2	1	2	0	-1	0	1	7	$7/1 = 7$
W	-3	-3	1	1	0	0	-15	

pivot

T1	x_1	x_2	e_1	e_2	a_1	a_2	RHS	
x_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	4	$L_1 \div 2$
a_2	0	$\frac{3}{2}$	$\frac{1}{2}$	-1	$-\frac{1}{2}$	1	3	$L_2 - L_1$
W	0	$-\frac{3}{2}$	$-\frac{1}{2}$	1	$\frac{3}{2}$	0	-3	$W + 3L_1$

1. Tie -3, -3: smallest index, x_1 enters.
2. Ratios: $8/2 = 4$, $7/1 = 7$.
3. Smallest 4: a_1 leaves; pivot 2.
4. T1: the point $(4, 0)$: $a_1 = 0$, but $a_2 = 3$ sensors missing. $W = -3$.



Phase 1, pivot 2: from $(4,0)$ to $(3,2)$

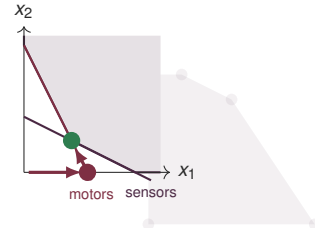
enters ↓

T1	x_1	x_2	e_1	e_2	a_1	a_2	RHS	ratio
x_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	4	$4/\frac{1}{2}=8$
leaves a_2	0	$\frac{3}{2}$	$\frac{1}{2}$	-1	$-\frac{1}{2}$	1	3	$3/\frac{3}{2}=2$
W	0	$-\frac{3}{2}$	$-\frac{1}{2}$	1	$\frac{3}{2}$	0	-3	

pivot

T2	x_1	x_2	e_1	e_2	a_1	a_2	RHS	
x_1	1	0	$-\frac{2}{3}$	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	3	$L_1-\frac{1}{2}L_2$
x_2	0	1	$\frac{1}{3}$	$-\frac{2}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	2	$L_2\div\frac{3}{2}$
W	0	0	0	0	1	1	0	$W+\frac{3}{2}L_2$

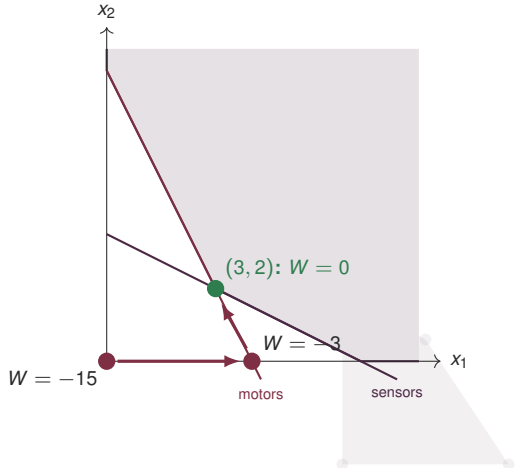
1. $-\frac{3}{2}$: x_2 enters.
2. Ratios: $4/\frac{1}{2}=8$, $3/\frac{3}{2}=2$.
3. Smallest 2: a_2 leaves; pivot $\frac{3}{2}$.
4. T2: no negative number: end of Phase 1. $W=0$: a real corner $(3,2)$.



The walk of Phase 1: from outside to a corner

Point	a_1 motors missing	a_2 sensors missing	W
O $(0,0)$	8	7	-15
$(4,0)$	0	3	-3
$(3,2)$	0	0	0

Phase 1 walks outside the feasible set, reducing what is missing, until it reaches a corner of the set.



Part 3 of 6 · minutes 34–43

Phase 2, and the two-phase method

1 Origin not a corner — 2 Phase 1 — **3 Phase 2** — 4 Example: minimum order — 5 No feasible plan — 6 Degeneracy, whole method

In this part, you learn to

- start Phase 2 from the corner found by Phase 1;
- write and clean the real objective line;
- read the answer, and the proof in the last line.

Phase 2: back to the real objective

end of Phase 1

T2	x_1	x_2	e_1	e_2	a_1	a_2	RHS
x_1	1	0	$-\frac{2}{3}$	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	3
x_2	0	1	$\frac{1}{3}$	$-\frac{2}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	2
W	0	0	0	0	1	1	0

delete

Ph. 2	x_1	x_2	e_1	e_2	RHS
x_1	1	0	$-\frac{2}{3}$	$\frac{1}{3}$	3
x_2	0	1	$\frac{1}{3}$	$-\frac{2}{3}$	2
z'	30	40	0	0	0

raw line

z'	0	0	$\frac{20}{3}$	$\frac{50}{3}$	-170

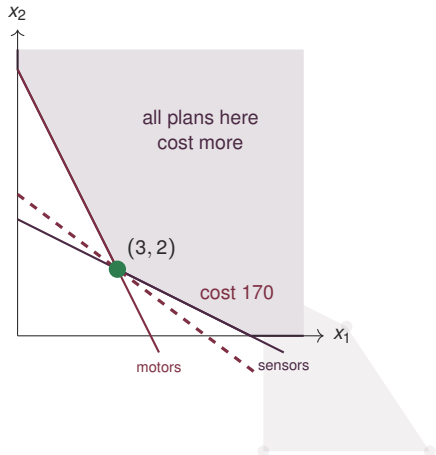
$z' - 30L_1 - 40L_2$

1. Delete the artificial columns: $a_1 = a_2 = 0$ must stay 0.
2. Real objective: $\min 30x_1 + 40x_2$, so $\max z' = -30x_1 - 40x_2$: line $z' + 30x_1 + 40x_2 = 0$.
3. x_1, x_2 are basic: clean the line (Rule 4.2).
4. No negative number: optimal. $z' = -170$: cost 170 at $(3, 2)$.

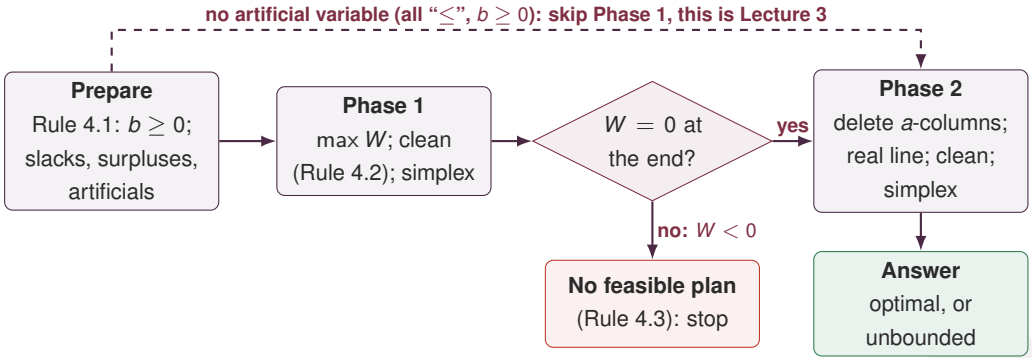
Reading the answer, and the proof

The last line reads $z' = -170 - \frac{20}{3}e_1 - \frac{50}{3}e_2$.

1. Cost $= -z' = 170 + \frac{20}{3}e_1 + \frac{50}{3}e_2$.
2. Every feasible plan has $e_1 \geq 0$, $e_2 \geq 0$ (extra motors, extra sensors).
3. So cost ≥ 170, and $(3, 2)$ costs exactly 170. ✓
4. $\frac{20}{3}$ and $\frac{50}{3}$ are the multipliers of Lecture 2 that proved "cost ≥ 170".



Method 4.1: the two-phase method



Part 4 of 6 · minutes 43–52

Example: a minimum order

1 Origin not a corner — 2 Phase 1 — 3 Phase 2 — **4 Example: minimum order** — 5 No feasible plan — 6 Degeneracy, whole method

In this part, you learn to

- solve a maximisation with a "$\geq$" constraint;
- follow Phase 1 and Phase 2 on the picture;
- read the surplus in the answer.

Example 4.2: a minimum order

A customer orders at least 12 drones B:

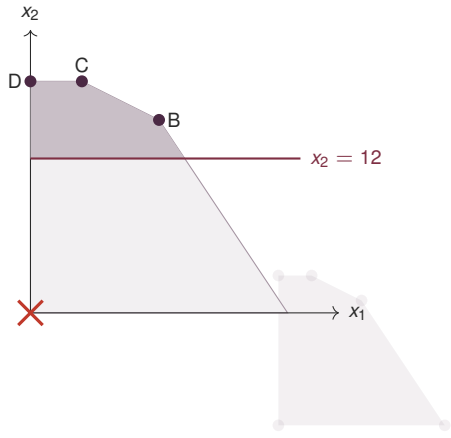
$$\max z = 40x_1 + 60x_2$$

$$\text{s.t. } x_1 + 2x_2 \leq 40, \quad 3x_1 + 2x_2 \leq 60, \quad x_2 \leq 18,$$

$$x_2 \geq 12, \quad x_1, x_2 \geq 0.$$

The origin is not feasible: $0 \geq 12$ is false.

Prepare (Rule 4.1): three slacks, and
$x_2 - e_4 + a_4 = 12$.
One artificial variable: Phase 1 is $\max\ W = -a_4$.



Example 4.2: Phase 1 in one pivot

enters ↓

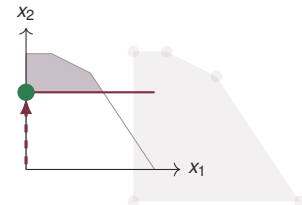
T0	x_1	x_2	s_1	s_2	s_3	e_4	a_4	RHS	ratio
s_1	1	2	1	0	0	0	0	40	$40/2 = 20$
s_2	3	2	0	1	0	0	0	60	$60/2 = 30$
s_3	0	1	0	0	1	0	0	18	$18/1 = 18$
leaves a_4	0	1	0	0	0	-1	1	12	$12/1 = 12$
W	0	-1	0	0	0	1	0	-12	

pivot

T1	x_1	x_2	s_1	s_2	s_3	e_4	a_4	RHS	
s_1	1	0	1	0	0	2	-2	16	$L_1 - 2L_4$
s_2	3	0	0	1	0	2	-2	36	$L_2 - 2L_4$
s_3	0	0	0	0	1	1	-1	6	$L_3 - L_4$
x_2	0	1	0	0	0	-1	1	12	$L_4 \div 1$
W	0	0	0	0	0	0	1	0	$W + L_4$

T0: W-line already cleaned ($W - L_4$).

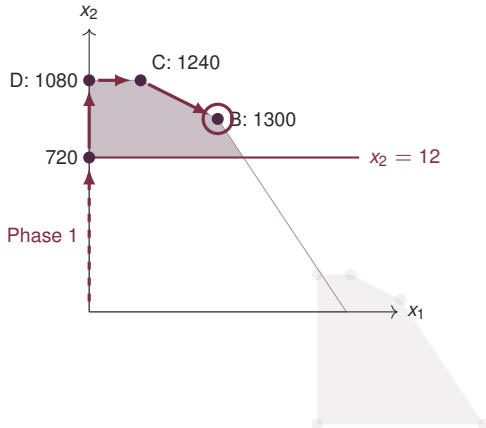
1. Only -1: x_2 enters.
2. Ratios 20, 30, 18, 12.
3. Smallest 12: a_4 leaves.
4. T1: $W = 0$: end of Phase 1, at $(0, 12)$.



Example 4.2: Phase 2, three pivots to B

Enters	Leaves	Corner	z
—	—	$(0, 12)$	720
e_4	s_3	D $(0, 18)$	1080
x_1	s_1	C $(4, 18)$	1240
s_3	s_2	B $(10, 15)$	1300

- e_4 enters: make more than 12 drones B.
- Last z-line $(0, 0, 25, 5, 0, 0)$: optimal, as in Lecture 2.
- $e_4 = 3$: 3 drones B more than ordered.



Part 5 of 6 · minutes 52–63

No feasible plan, and special cases

1 Origin not a corner
2 Phase 1
3 Phase 2
4 Example: minimum order
5 **No feasible plan**
6 Degeneracy, whole method

In this part, you learn to

- recognise a problem with no feasible plan from Phase 1 (Rule 4.3);
- treat equalities and negative right-hand sides;
- handle an artificial variable still basic at the end of Phase 1.

Signal 3: Phase 1 ends with $W < 0$

max $x_1 + x_2$ s.t. $x_1 + x_2 \le 1$, $x_1 + x_2 \ge 2$, $x \ge 0$ (Lecture 2, Case 5). W-line cleaned.

enters ↓

	T0	x_1	x_2	s_1	e_2	a_2	RHS	ratio
leaves	s_1	1	1	1	0	0	1	$1/1 = 1$
	a_2	1	1	0	-1	1	2	$2/1 = 2$
	W	-1	-1	0	1	0	-2	

pivot

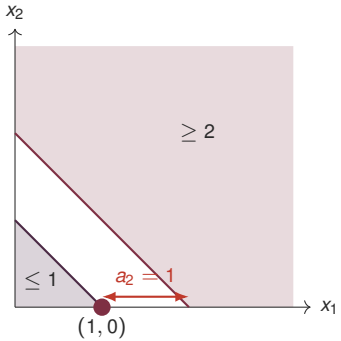
T1	x_1	x_2	s_1	e_2	a_2	RHS	
x_1	1	1	1	0	0	1	$L_1 \div 1$
a_2	0	0	-1	-1	1	1	$L_2 - L_1$
W	0	0	1	1	0	-1	$W + L_1$

1. Tie -1, -1: x_1 enters.
2. Ratios 1 and 2.
3. Smallest 1: s_1 leaves.
4. T1: no negative number: Phase 1 ends, but $W = -1 < 0$.

Rule 4.3 (no feasible plan)

Phase 1 ends with $W < 0 \Rightarrow$ no feasible plan (Key fact 4.1). Stop.

What $W < 0$ tells us



- The two half-planes do not meet.
- Phase 1 stops at $(1, 0)$: the best it can do.
- There $a_2 = 1$: we still miss 1 unit in the line $x_1 + x_2 \geq 2$. This is the gap between the two lines.
- In practice: an error in the data, or needs too high for the resources.

Equalities and negative right-hand sides

- **Negative b:** $x_1 - 2x_2 \leq -2$
  $\rightarrow$ multiply by -1: $-x_1 + 2x_2 \geq 2$
  $\rightarrow$ $-x_1 + 2x_2 - e + a = 2$.
- **Equality:** $x_1 + x_2 + x_3 = 6$
  $\rightarrow$ $x_1 + x_2 + x_3 + a = 6$ (one artificial variable).
- **Why $b \geq 0$ first?** The starting basic variables are equal to b, and they must be ≥ 0.

Do not

- add an artificial variable to a "$\leq$" line with $b \geq 0$: its slack is enough;
- write a surplus with a "$+$" sign;
- replace "$=$" by two inequalities here: one artificial variable is simpler.

A special end of Phase 1

Sometimes Phase 1 ends with $W = 0$, but an artificial variable is still basic, with the value 0.

- The plan is real (all $a = 0$), but this artificial variable must leave before Phase 2.
- Its line has RHS $= 0$. Choose a non-zero number of this line under a non-artificial column, and pivot on it. No ratio test: the RHS is 0, so no value changes.
- If this line has only zeros under the non-artificial columns, the constraint was redundant: delete the line.

Remember only $W < 0$ means "no feasible plan". $W = 0$ always gives a real plan.

Part 6 of 6 · minutes 63–88

Degeneracy, and the complete method

1 Origin not a corner — 2 Phase 1 — 3 Phase 2 — 4 Example: minimum order — 5 No feasible plan — 6 **Degeneracy, whole method**

In this part, you learn to

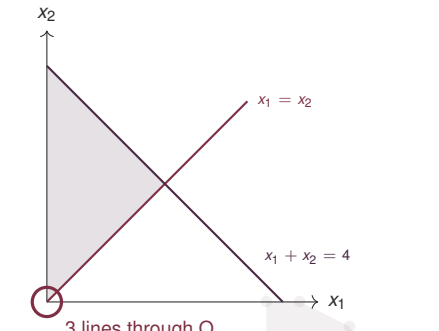
- recognise a degenerate corner and a degenerate pivot;
- know Bland's rule, which prevents cycling;
- see the complete simplex method and all its signals.

Degeneracy: a basic variable equal to 0

Definition 4.2

A basic feasible solution is degenerate when at least one *basic* variable is equal to 0.

- It comes from a tie in the ratio test, or from a right-hand side $b_i = 0$.
- On the picture: more lines than necessary pass through the corner (Lecture 2, special situation B).
- Then the next ratio can be 0: the pivot changes the basis, but not the point.



$\max\ 2x_1 + x_2,\ x_1 - x_2 \leq 0,\ x_1 + x_2 \leq 4$

A degenerate pivot: ratio 0

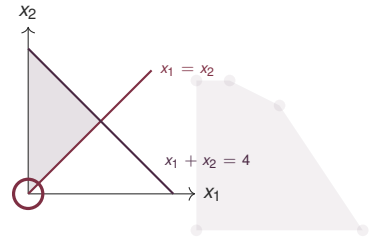
enters ↓

	T0	x_1	x_2	s_1	s_2	RHS	ratio
leaves	s_1	1	-1	1	0	0	$0/1 = 0$
	s_2	1	1	0	1	4	$4/1 = 4$
	z	-2	-1	0	0	0	

pivot ↓

T1	x_1	x_2	s_1	s_2	RHS	
x_1	1	-1	1	0	0	$L_1 \div 1$
s_2	0	2	-1	1	4	$L_2 - L_1$
z	0	-3	2	0	0	$z + 2L_1$

1. -2: x_1 enters.
2. Ratios: $0/1 = 0$, $4/1 = 4$.
3. Smallest 0: s_1 leaves; x_1 enters with the value 0.
4. T1: still $(0, 0)$, $z = 0$: same point, new basis.



Then a useful pivot

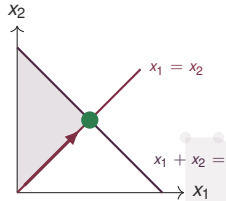
enters ↓

	T1	x_1	x_2	s_1	s_2	RHS	ratio
	x_1	1	-1	1	0	0	-1: no ratio
leaves	s_2	0	2	-1	1	4	$4/2 = 2$
	z	0	-3	2	0	0	

pivot

T2	x_1	x_2	s_1	s_2	RHS	
x_1	1	0	$\frac{1}{2}$	$\frac{1}{2}$	2	$L_1 + L_2$
x_2	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	2	$L_2 \div 2$
z	0	0	$\frac{1}{2}$	$\frac{3}{2}$	6	$z + 3L_2$

1. -3: x_2 enters.
2. Line x_1 has -1: no ratio; $4/2 = 2$.
3. s_2 leaves; pivot 2.
4. T2: optimal, $(2, 2)$, $z = 6$.



The degenerate pivot chose a description of O from which we can move.

Cycling, and Bland's rule

- With degenerate pivots, z may stay the same for several pivots.
- In rare, specially built examples (E. Beale, 1955), the method comes back to a basis already visited and turns in a circle forever: cycling.
- It almost never happens in practice, but a simple rule makes it impossible.

Key fact 4.2 (Bland's rule; admitted)

Entering variable: the first variable (smallest index) with a negative number in the objective line. Leaving variable: among the smallest ratios, the line whose basic variable has the smallest index. Then the simplex method always stops.

In this course: the most negative number, ties by smallest index. If you see the same z for several pivots, switch to Bland's rule.

Other books: the Big-M method

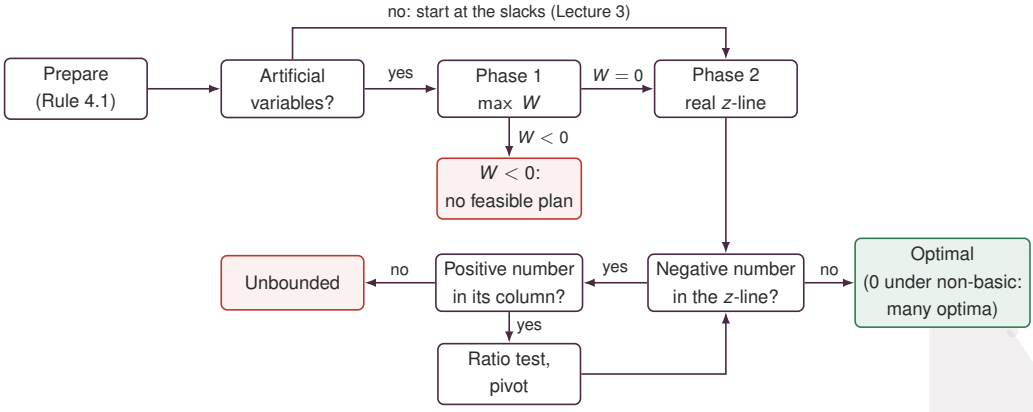
- One single phase: the artificial variables go into the real objective with a huge penalty M:

$$\max\ z = -30x_1 - 40x_2 - Ma_1 - Ma_2 \qquad (M \text{ “very large”}).$$

- The penalty pushes the artificial variables to 0, as Phase 1 does.
- But the objective line contains M: compare $30 - 3M$ and $40 - 3M$, or choose a value of M (rounding errors on a computer).

In this course we use the two-phase method: no huge numbers, and a clear signal $W < 0$.

The complete simplex method



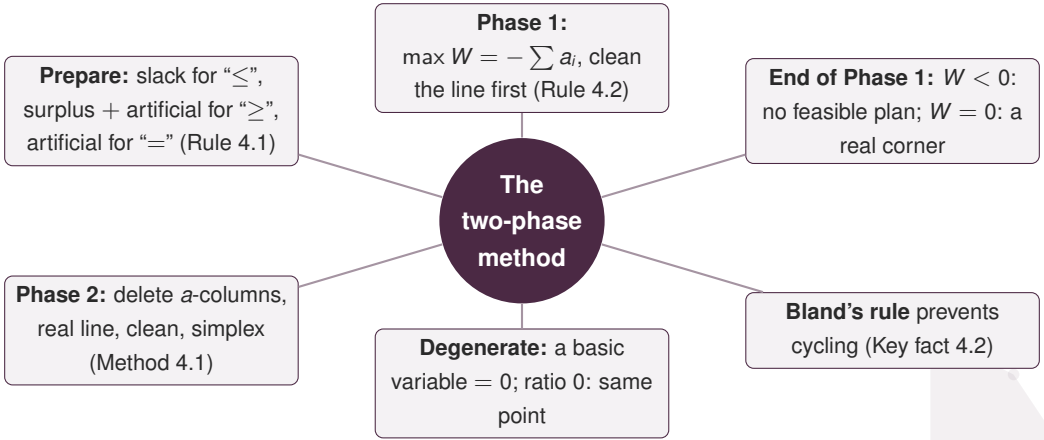
All the signals in one table

Where	What we see	Conclusion
End of Phase 1	$W < 0$	no feasible plan (Rule 4.3)
End of Phase 1	$W = 0$	a real corner: go to Phase 2
Objective line	no negative number	optimal (Key fact 3.2)
Objective line	a 0 under a non-basic variable	many optimal plans (Rule 3.4)
Entering column	no positive number	unbounded (Rule 3.3)
Ratio test	smallest ratio $= 0$	degenerate pivot: same point

Common mistakes

Wrong $\times$	Right $\checkmark$
Forget to clean the W-line or the z-line	Clean before the first pivot of each phase (Rule 4.2)
Add the variables before making $b \geq 0$	First multiply by -1 the lines with $b < 0$
An artificial variable on a "$\leq$" line	Only on "$\geq$" and "$=$" lines
Keep the artificial columns in Phase 2	Delete them at the end of Phase 1
Read $z' = -170$ as a cost of -170	For a minimisation: cost $= -z' = 170$
"$W = 0$ with an a still basic: infeasible"	Only $W < 0$ means no feasible plan

Summary: the two-phase method in one picture



TD 3: the two-phase method by hand. Next: the meaning of the last z-line (Chapter 3).